THE NEARBY-OPEN QUASI-SECTION CONCLUSION SURVIVES REPLACING QUASI-FINITENESS BY LOCAL FINITE TYPE PLUS A CLOSED POINT IN THE FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 15053-15061 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 15063-15075, the complete author proof (11 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Companion to the already-collected sibling meeting-nearby-open (L16081); the reduction here is complete, the cited lemma's conclusions are carried by reference as intended by the author.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The chapter's lemma quasi-finite-quasi-section-meeting-nearby-open-X (the source of the conclusions; queued as the next wave's card for the X-version), Noetherianity of finite type schemes over locally Noetherian bases, quasi-compactness of opens in Noetherian spaces, and the scheme-theoretic-image specialization fact.

Source reference. Original lemma 05GU: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14994–15004. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01T6: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 15075–15075. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02JQ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 15075–15075. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0052: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 15075–15075. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04ZA: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 15075–15075. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Suppose that $f : X \to S$ is locally of finite type, S locally Noetherian, $x \in X$ a closed point of its fibre X_s, and $U \subset X$ an open subscheme such that $U \cap X_s = \emptyset$ and $x \in \overline{U}$, then the conclusions of Lemma 05GU hold.*

Proof. Namely, we can reduce this to the cited lemma as follows: First we replace X and S by affine neighbourhoods of x and s. Then X is Noetherian, in particular U is quasi-compact (see Morphisms, Lemma 01T6 and Topology, Lemmas 0052 and 04ZA). Hence there exists a specialization $x' \rightsquigarrow x$ with $x' \in U$ (see Morphisms, Lemma 02JQ). Note that $f(x') \neq s$. Thus we see all hypotheses of the lemma are satisfied and we win. □